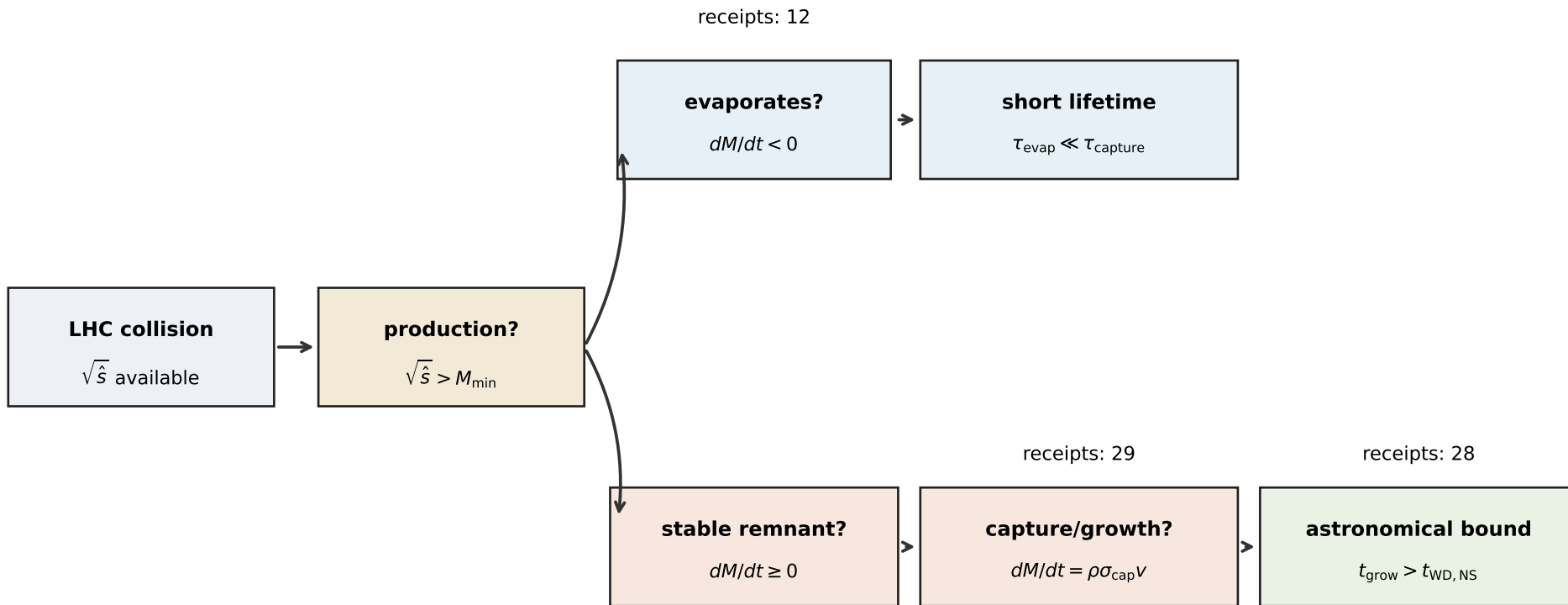


# Branched physical tree for the LHC black-hole catastrophe claim



The catastrophe branch requires production, survival, capture, rapid growth, and failure of the astronomical bound.